

Git & GitHub Command Guide

A Complete Reference for Version Control Mastery

Basic Configuration

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```

Sets up your identity for all local repositories

Repository Initialization

```
git init                # Initialize a new local repository
git clone <url>          # Clone a remote repository
```

Basic Git Operations

Staging Changes

```
git status              # Check working directory status
git add <file>          # Add specific file to staging
git add .               # Add all changes to staging
git reset <file>        # Unstage changes for specific file
```

Committing Changes

```
git commit -m "message" # Commit staged changes
git commit -am "message" # Stage tracked files and commit
```

Branch Management

```
git branch                # List branches
git branch <branch-name>  # Create new branch
git checkout <branch-name> # Switch to branch
git checkout -b <new-branch> # Create and switch to new branch
git branch -d <branch-name> # Delete branch
```

Remote Operations

```
git remote add origin <url>    # Add remote repository
git push origin <branch>        # Push changes to remote
git pull origin <branch>        # Pull changes from remote
git fetch                       # Download objects from remote
```

Advanced Operations

Stashing

```
git stash                # Temporarily store changes
git stash list           # List stashed changes
git stash pop            # Apply and remove stashed changes
git stash drop           # Delete stashed changes
```

History and Differences

```
git log                  # View commit history
git log --oneline        # Compact commit history
git diff                 # Show unstaged changes
git diff --staged        # Show staged changes
```

Merging and Rebasing

```
git merge <branch>       # Merge branch into current branch
git rebase <branch>      # Rebase current branch onto another
git merge --abort         # Abort troubled merge
```

Recovery and Reset

```
git checkout -- <file>      # Discard changes in working directory
git reset --hard HEAD       # Reset to last commit
git revert <commit>         # Create new commit that undoes changes
```

GitHub Specific Operations

```
# Fork Management
git remote add upstream <original-repo-url> # Add original repo as upstream
git fetch upstream                          # Fetch upstream changes
git merge upstream/main                    # Merge upstream changes

# Pull Requests
git push origin <branch> # Push branch for PR
```

Best Practices

1. Commit Messages

- Use present tense ("Add feature" not "Added feature")
- Be descriptive but concise
- Start with a capital letter

2. Branching Strategy

- main/master: stable production code
- develop: integration branch
- feature/: new features
- hotfix/: urgent fixes

3. Pull Request Guidelines

- Keep changes focused and small
- Include clear descriptions
- Reference related issues

Common Workflows

Feature Development

```
git checkout -b feature/new-feature
git add .
git commit -m "Add new feature"
git push origin feature/new-feature
# Create PR on GitHub
```

Hotfix Process

```
git checkout -b hotfix/bug-fix main
git add .
git commit -m "Fix critical bug"
git push origin hotfix/bug-fix
# Create PR on GitHub
```